

SAI RAHUL BHAT GAADHI RAJU

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PROFESSIONAL SUMMARY

B.Tech Electronics and Communication Engineering student specializing in RTL design, digital systems, and analog/mixed-signal VLSI design flows. Samsung ISWDP Fellow with hands-on training in semiconductor fundamentals, CMOS device physics, and TCAD-based modeling. Experienced in designing, simulating, and verifying digital and analog IC architectures using Verilog HDL, Cadence Virtuoso, Xilinx Vivado, and Synopsys Sentaurus TCAD. Seeking a core industry internship in RTL design, functional verification, or analog/physical design starting December/January.

EDUCATION

Anurag University, Hyderabad, India

Bachelor of Technology in Electronics and Communication Engineering (ECE) — Expected Graduation: 2027

- Cumulative Grade Point Average: 7.95 / 10.0

Jawaharlal Nehru Government Polytechnic (JNGPT), Ramanthapur, India

Diploma in Electronics and Communication Engineering (ECE) — 2021 – 2024

- Cumulative Grade Point Average: 9.17 / 10.0 (Graduated with Distinction)

TECHNICAL TRAINING & INDUSTRY EXPOSURE

Samsung Fellowship — India Semiconductor Workforce Development Program (ISWDP) | Cohort 8 Scholar, Grade II

- Selected from a national pool of 4,000+ engineering applicants based on technical aptitude.
- Completed industry-led training in deep-submicron CMOS fundamentals, short-channel effects, and physical layout principles.
- Built active device physics models in Synopsys Sentaurus TCAD to study switching characteristics, threshold voltage variation, and transistor-level behavior.

TECHNICAL SKILLS

- Hardware Description Languages: Verilog HDL (RTL, Behavioral, Dataflow, Structural), Verilog-A
- EDA Tools: Cadence Virtuoso, Synopsys Sentaurus TCAD, Xilinx Vivado, ModelSim
- Simulation Tools: Spectre (AC, Transient, Mixed-Signal)
- Analog & Mixed-Signal Design: Switched-Capacitor Circuits, Operational Amplifier Design, Comparators, Sigma-Delta ADC, Noise Shaping, Low-Power Analog Design
- Digital Design: RTL Design, Combinational & Sequential Logic, Finite State Machines (FSM)
- FPGA Development: FPGA Prototyping, Constraint Definition, Pin Mapping, Clock Definition
- Verification: Testbench Development, Waveform Debugging, Simulation-Based Functional Verification
- Semiconductor Concepts: CMOS, FinFET, Static Timing Analysis (STA), Setup/Hold Constraint Validation
- Programming & Scripting: C, Python (Automation & Scripting), MATLAB (DSP & Filter Design)

CORE VLSI & DIGITAL PROJECTS

Design of a Low-Power 2nd-Order Sigma-Delta ADC Using Noise Shaping Techniques | Cadence Virtuoso, Spectre | 18nm FinFET

- Designed a second-order Sigma-Delta ADC architecture using switched-capacitor integrators to achieve in-band noise shaping.
- Designed a two-stage Miller-compensated operational amplifier as the core analog building block for the integrator stages.
- Developed the comparator and 1-bit feedback DAC completing the ADC feedback loop.
- Implemented a Sinc³ (CIC) decimation filter in Verilog-A for digital output filtering.
- Performed AC, transient, and mixed-signal simulations in Cadence Virtuoso and Spectre on 18nm FinFET technology.
- Optimized the analog core for low-power operation, achieving approximately 51.36 μ W power consumption.

RISC-V 4-Stage Pipelined ALU Design | Xilinx Vivado

- Designed a 4-stage pipelined Arithmetic Logic Unit (Fetch, Decode, Execute, Writeback) aligned with RV32I ISA semantics, separating datapath and control logic in Verilog HDL.
- Implemented data-forwarding paths and pipeline stall logic to address structural and data hazards across sequential instructions.
- Simulated and functionally verified the design in Xilinx Vivado using testbench-driven verification across multiple instruction sequences.

FPGA-Based Power Quality Anomaly Detection Core | Xilinx Vivado

- Developed a Verilog HDL hardware core to detect voltage fluctuations and frequency-based power quality anomalies.

- Synthesized the design in Xilinx Vivado, applying hardware constraints, pin mapping, and clock definitions for target FPGA implementation.
- Validated functional behavior and timing through testbench-driven simulation.

16-Bit ALU Comparative Study (Pipelined vs. Non-Pipelined) | *ModelSim & Vivado*

- Developed a Verilog HDL test framework to compare pipelined and non-pipelined ALU datapath architectures.
- Synthesized both design variants to evaluate VLSI performance metrics, including maximum clock frequency (Fmax), propagation delay, and resource utilization.
- Analyzed timing behavior and architectural trade-offs between the two implementations through waveform debugging in ModelSim.

Parameterized N-bit Barrel Shifter Architecture | *Xilinx Vivado*

- Designed a parameterized N-bit barrel shifter using a cascaded 2-to-1 multiplexer tree ($\log_2 N$ stages) supporting logical and arithmetic shift operations.
- Implemented the RTL in Verilog HDL, structuring combinational logic to ensure clean synthesis with no inferred latches.
- Verified functional correctness across multiple test vectors and control-word combinations through simulation.

CERTIFICATIONS

Semiconductor & VLSI — NPTEL

- VLSI Physical Design with Timing Analysis — IIT Roorkee
- Digital Design with Verilog HDL — IIT Guwahati (Top 75th percentile)
- VLSI Design Flow: RTL to GDS — IIIT Delhi

Semiconductor & VLSI — Maven Silicon

- VLSI SoC Architecture & Digital Design Overview